

# NAZIRU HALILU

*AI · Computer Vision · Geospatial ML for Digital Agriculture*

Pamplona, Spain | +34 6 5296 7216 | [halilu.175366@e.unavarra.es](mailto:halilu.175366@e.unavarra.es)

[GitHub](#) | [ORCID](#) | [Zenodo records](#) | [Portfolio](#)

Agricultural engineer with a long-term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high-throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems. Current interests include self-supervised deep learning for crop-stress detection and vegetation-index estimation from UAV and multispectral imagery, and closed-loop sensing and control for irrigation, greenhouse automation, and precision farm machinery.

*This document is a complete, single-file capture of every section of the live portfolio site ([halilunaziru73-creator.github.io](https://halilunaziru73-creator.github.io)), including all gallery photographs and certificate images, generated for offline reading and archival purposes.*

## EDUCATION

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- Erasmus Mundus Joint Master's Degree (EMJMD), Cutting-Edge Technologies for Sustainable Agriculture** 2025 to expected 2027  
*Public University of Navarre, Spain · UTAD, Portugal · Agricultural University of Athens, Greece*
- Bachelor of Engineering (B.Eng.), Agricultural and Bioresources Engineering, First Class Honours** 2019 to 2024  
*Ahmadu Bello University, Zaria, Nigeria · CGPA 4.77/5.00, top 1% of graduating cohort*
- National Diploma, Agricultural Engineering Technology, Distinction** 2016 to 2018  
*Ahmadu Bello University, Zaria, Nigeria · CGPA 3.78/4.00, top 1% of graduating cohort*

## RESEARCH EXPERIENCE

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- EMJMD Research Scholar (Mobility Track)** Sep 2026 to Jan 2027  
*Agricultural University of Athens, Greece*
- Engineered NaCROP, a modular Python irrigation and crop-growth decision-support pipeline with a Tkinter GUI, validated across five smallholder crops around Zaria, Nigeria. [Code: GitHub](#)
- Summer Intern, Computer Vision Group** Jun 2026 to Aug 2026  
*Dept. of Computer Engineering, Ahmadu Bello University, Zaria*
- Extended the GACL framework with four new geometry-invariant components, reaching 64.4% accuracy (macro AUC 0.882) across 19 disease classes. [Code: GitHub](#)
- EMJMD Research Scholar (Mobility Track)** Feb 2026 to Jun 2026  
*University of Trás-os-Montes and Alto Douro (UTAD), Portugal*
- Integrated GIS, remote sensing, and ML pipelines across three cropping systems (R-squared 0.978–0.995), cutting lime demand by 2.69 tonnes. [Code: GitHub](#)
  - Developed Multitask Singularity Regression Kriging (MSRK), reducing residual kriging uncertainty by roughly 20% across 17,864 soil samples. [Code: GitHub](#)
  - Built an NDVI/logarithmic regression model forecasting grape ripening (R-squared = 0.9784) across a 6-hectare UTAD vineyard.
  - Ran terrain characterisation and a 25-trap spatial monitoring network for grape moth across a 42.97-hectare Douro vineyard estate.
- EMJMD Research Scholar (Mobility Track)** Sep 2025 to Jan 2026  
*Public University of Navarre, Spain*
- Built a 14-module open-source pipeline for RGB/multispectral vegetation indexing, GLCM texture analysis, and CNN/Faster R-CNN inference, reaching 33.8% balanced accuracy against a 4.55% chance baseline. [Code: GitHub](#)
  - Designed STADOS, a five-stage radiometric harmonization framework, cutting temporal NDVI variance by 39% and 31% on Sentinel-2 imagery. [Code: GitHub](#)
  - Built a Bayesian digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models for gully erosion biocontrol (hold-out R-squared = 0.90). [Code: GitHub](#)
- Undergraduate Project Assistantships** Mar 2024 to Mar 2025  
*Depts. of Agricultural and Bioresources Engineering and Animal Science, ABU*
- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects spanning irrigation, soil erosion, postharvest, and animal systems.
- Project Assistant (Remote Data Lead), Smart Greenhouse Automation** 2024 to 2025  
*IAR-ABU, Nigeria*
- Automated greenhouse environmental control by engineering adaptive irrigation-scheduling algorithms in Python.
- Student Research Assistant, Precision Crop Systems** 2022 to 2024  
*NAERLS, Nigeria*
- Trained CNN-based crop-stress detection models in TensorFlow, reaching 78% classification accuracy on multispectral UAV imagery.
- Assistant Farm Manager (concurrent, part-time)** 2022 to 2024  
*Yankari Nasarawa Farms, Nigeria*

- Managed automated climate-control and water-dosing systems across a commercial poultry facility.

### **Project Assistant, Smart Irrigation and Water Systems**

**2021 to 2022**

*BOMO-Basin Project, Nigeria*

- Automated water delivery across a one-hectare smart drip-irrigation array via closed-loop control logic.

### **Research Intern**

**2017 to 2020**

*NCAM, Ilorin and IAR, Zaria, Nigeria*

- Built Python data-logging pipelines benchmarking mechanization efficiency; supported baseline hydrological research.

### **Founder, Nhalilu's Biogas Plant and Sales Company**

*Registered with SMEDAN, Bioprocess Engineering*

- Founded and registered a biogas production and sales venture applying bioprocess engineering principles.

## PUBLICATIONS

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### Peer-Reviewed

- **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *IJRSI*, 13(7), 2920–2950. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951–2989. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. *IJRSI*, 13(7), 2990–3020. [DOI](#)
- Ismail, K., **Halilu, N.** (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. AfCPA*, 2(1), 100–120.
- Auwal, M., Yusuf, A., **Halilu, N.** (2020). Physio-chemical and bacteriological quality of water for irrigation purpose (Shika Dam). *Nigerian Research Journal of Engineering and Environmental Sciences*, 5(1), 198–204.

### Submitted / Under Review

- **Halilu, N.,** Sulaiman, J. Geometry-Agnostic Contrastive Learning (GACL). *Biosystems Engineering* (Elsevier). Under review. [Code](#)
- **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory. *Environmental Monitoring and Assessment*. Under review. [Code](#)
- **Halilu, N.,** Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy*. Under review. [Code](#)

### Preprints

- **Halilu, N.** (2026). Multitask Singularity Regression Kriging. *Zenodo Preprint*. [Code](#)
- **Halilu, N.** (2026). STADOS: Spectral-Temporal Adaptive Dark Object Subtraction. *Zenodo*. [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time-Series Modelling and Control Systems in Precision Irrigation. *Zenodo*.

## RESEARCH PROJECTS AND OPEN-SOURCE REPOSITORIES

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- Geometry-Agnostic Contrastive Learning (GACL): <https://github.com/halilunaziru73-creator/Geometry-Agnostic-Contrastive-Learning-GACL>
- STADOS: Spectral-Temporal Adaptive Dark Object Subtraction: <https://github.com/halilunaziru73-creator/STADOS-Spectral-Temporal-Adaptive-Dark-Object-Subtraction>
- Multitask Singularity Regression Kriging (MSRK): <https://github.com/halilunaziru73-creator/Multitask-Singularity-Regression-Kriging-for-Spatio-Temporal-Soil-NPK-Crop-Stress-Yield-Prediction>
- Digital Twin for Gully Biocontrol Using Morning Glory: <https://github.com/halilunaziru73-creator/Digital-Twin-for-the-Evaluation-of-Experimental-Gully-Biocontrol-Using-Morning-Glory-Ipomoea-spp>
- Hybrid CNN-BiLSTM-Attention Sediment Transport Framework: <https://github.com/halilunaziru73-creator/Hybrid-CNN-BiLSTM-Attention-Sediment-Transport-Agricultural-Gully-System>
- Real-Time RGB Proxy Vegetation Indexing and Texture Analysis: <https://github.com/halilunaziru73-creator/Real-Time-RGB-Proxy-Vegetation-Indexing-and-Texture-Analysis-for-UAV-and-Handheld-Crop-Imagery>
- Operationalizing GIS and ML across Contrasting Cropping Systems: <https://github.com/halilunaziru73-creator/Operationalizing-GIS-and-Machine-Learning-across-Contrasting-Cropping-Systems>
- Geospatial Data Analysis (six-notebook GIS course): <https://github.com/halilunaziru73-creator/Geospatial-data-analysis>
- GIS-Based Livestock Slurry Suitability Delineation: <https://github.com/halilunaziru73-creator/GIS-based-delineation-of-areas-suitable-for-livestock-slurry-application>
- GIS Monitoring of Grape Moth (*Lobesia botrana*): <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Grape-Moth-Vineyard-Pest>

- GIS Monitoring of Vineyard Ripening (Brix): <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Vineyard-Ripening-Brix>
- NaCROP: Crop ET and Irrigation Decision-Support Pipeline: <https://github.com/halilunaziru73-creator/NaCROP>
- Integrated Decision-Support Model for Dairy Processing: <https://github.com/halilunaziru73-creator/Integrated-Decision-Support-Model-for-Dairy-Processing-under-Milk-and-Market-Variability>
- Full repository index: <https://github.com/halilunaziru73-creator>

## SKILLS

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**Geospatial AI / GIS:** QGIS, ArcGIS, ENVI, Google Earth Engine, Sentinel imagery, UAV imaging

**Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, CNNs, self-supervised learning

**Crop Modelling:** APSIM Next Generation, crop growth simulation

**Climate AI:** CMIP6, ERA5, ACCESS-S2, downscaling and forecasting

**3D Computer Vision:** COLMAP, Open3D, Pix4D, Agisoft Metashape

**Programming:** Python, MATLAB, R, Git/GitHub, Zenodo

## TEACHING, LEADERSHIP AND COMMUNITY

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### Undergraduate Course Tutor

2020 to 2024

*ABU Zaria (NAMET)*

- Ran supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agricultural Engineering.

### Secretary General, NAAESB

2023 to 2024

*National Assoc. of Agric. Engineering Students*

- Directed operational targets and inter-chapter scientific meetups for the national student body.

### Deputy Coordinator

2020 to 2025

*KOJYAF, Zango Shanu, Zaria*

- Coordinated tutorial and mentorship programmes preparing secondary students for JAMB entrance examinations.

### Director, Environmental Protection Agency

2023 to 2024

*ABU Students' Union Government*

- Directed campus environmental protection and sanitation initiatives.

### Physics and Mathematics Teacher

2022 to 2023

*Takai Academy, Zaria, Nigeria*

- Instructed senior classes in classical physics and computational principles.

## AWARDS AND HONOURS

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- Erasmus Mundus Joint Master Scholarship, European Union (2025)
- Best Project Award, ABU Senate (2024)
- Best Graduating Student, Agricultural and Bioresources Engineering (2024)
- Nigerian Federal Government Award for Academic Excellence (2024)
- Best Undergraduate Research Paper Award, ABU (2024)
- Best 300 Level Student, ABU (2022)
- Best 200 Level Student, ABU (2021)
- Award of Merit as Tutor, Faculty of Engineering, ABU (2021)
- Best Graduating Student, National Diploma, Samaru College of Agriculture (2019)
- PTDF Undergraduate Fellow (2023)
- Needs-Based University Fellowship (2022)
- Award of Honour, International Career Coaching Program, Envirofly Consulting under UK FCDO (2024)

- Award for Best Graduating Student, National Union for Engineering Student Association (2024)
- Award of Merit, Al-Azhar Institute of Arabic and Islamic Studies (2022)
- Award of Merit as Senator, Student Union Government, Samaru College of Agriculture (2017)
- Best Student in Science Subjects, Government Secondary School Jama'a (2014)

## **CONFERENCES, SEMINARS AND WORKSHOPS**

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- Data Management Workshop: From Phenotyping Experiment to FAIR Data, Norway (2026)
- Digital Agroecology Cluster Conference (EU-funded), Brussels (2025)
- 6th African Regional Conference on Irrigation and Drainage, ARCID (2025)
- Vibrational Spectroscopy and Chemometrics in Agro-Food Systems (EU-funded) (2025)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, University of Florida (2025)
- IEEE Intl. Conf. on Smart Agriculture Technologies (AgriTech), Valencia, Spain (2024)
- NAERLS Technical Seminar Series: AI in the Food Value Chain (2023)

## **PROFESSIONAL MEMBERSHIPS**

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American Society of Agricultural and Biological Engineers (ASABE) · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN

## REFEREES

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**Prof. Javier Casalí Sarasibar**, Public University of Navarre, Spain · [jcs@unavarra.es](mailto:jcs@unavarra.es)

**Prof. Konstantinos Soulis**, Agricultural University of Athens, Greece · [soco@aua.gr](mailto:soco@aua.gr)

**Prof. Miguel Ángel Campo Bescós**, Public University of Navarre, Spain · [miguel.campo@unavarra.es](mailto:miguel.campo@unavarra.es)

**Prof. José Tadeu Marques Aranha**, UTAD, Portugal · [j\\_aranha@utad.pt](mailto:j_aranha@utad.pt)

**Dr. Ibrahim Babangida Dalha**, Ahmadu Bello University, Nigeria · [ibdalha@abu.edu.ng](mailto:ibdalha@abu.edu.ng)

**Dr. Habibu Ismail**, Ahmadu Bello University, Nigeria · [habfta@yahoo.com](mailto:habfta@yahoo.com)

**Dr. Mohammed Abdulsalam**, Ahmadu Bello University, Nigeria · [mabdulsalam@abu.edu.ng](mailto:mabdulsalam@abu.edu.ng)

**Dr. Yahya Gambo**, KFUPM, Saudi Arabia · [yahya.Gambo@kfupm.edu.sa](mailto:yahya.Gambo@kfupm.edu.sa)

# CERTIFICATIONS AND DOCUMENTED RECOGNITIONS

## Honorary Awards



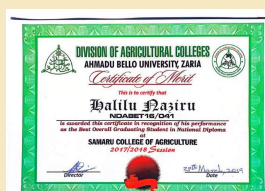
**Award of Merit, Senator (SRC/SUG)**

Samaru College of Agriculture  
SUG, 2017/2018 session



**Certificate of Merit, Best 200 Level Student**

Dept. of Agric. & Bio-Resources  
Engineering, ABU, 2018/2019  
session



**Best Overall Graduating Student, National Diploma**

Samaru College of Agriculture,  
ABU, 2017/2018 session



**Best Graduating Student, ND Agric. Engineering Technology**

Samaru College of Agriculture,  
ABU, 2017/2018 session

## Training



**Certificate of Continuing Professional Development**

IICCP, Envirofly Consulting (UK),  
60 CPD hours, 2024



**Tony Elumelu Foundation Certificate**

Business Management Training,  
2020



**Certificate of Attendance, Explore AI in Bioinformatics**

Genomac Institute Inc., USA, 2025



**Certificate of Learning, Genomics and Bioinformatics Bootcamp**

Genomac Institute Inc., USA, 2024



**International Webinar Series for Life Scientists**

Genomac Institute Inc., USA, 2024



**Bioinformatics Course 201**

Genomac Institute Inc., USA, 2024



**Genomics and Bioinformatics Short Course 301**

Genomac Institute Inc., USA, 2024



**SIWES Industrial Attachment Attestation**

NAERLS, ABU, 2023



**SIWES Industrial Attachment Attestation**

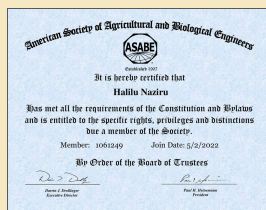
IAR, Samaru, ABU, 2017/2018



**Mechanistic Modeling Seminar Attendance**

ISFOOD Institute, UPNA, 2025

## Recognition



**ASABE Member Certificate**  
Member No. 1061249, joined 2022



**NIAE Student Member Certificate**  
Membership No. S2644, 2024



**Conference Paper Acceptance Letter**  
6th ARCID 2025, Abuja



**IJRSI Publication Certificate**  
Sediment Transport paper, Vol. 13  
Issue 7, 2026



**IJRSI Publication Certificate**  
Real-Time RGB Vegetation  
Indexing paper, Vol. 13 Issue 7,  
2026



**IJRSI Publication Certificate**  
NIR/Hyperspectral review paper,  
Vol. 13 Issue 7, 2026



**SMEDAN Certificate of Registration**  
N.halilus Biogas Plant & Sells,  
2020



**Attestation to Proficiency in English Language**  
Ahmadu Bello University, 2024

## West African Examinations Council



**West African Senior School Certificate (WAEC)**  
Knowledge Is Power Secondary  
School, Zaria, 2015

## FULL ACADEMIC RECORD

Course-by-course record across National Diploma, B.Eng., and EMJMD programmes. Full authenticated transcripts are legalized by the Federal Ministry of Education, Nigeria, and the Consulado General de España en Lagos.

### ND Grading Scale

| Score %  | Letter Grade (LG) | Weighted Points (WP) | CGPA/Grade of Certificate |   |
|----------|-------------------|----------------------|---------------------------|---|
| 80 – 100 | A                 | 4                    | 3.5 – 4.0 Distinction     | A |
| 70 – 79  | AB                | 3.5                  | 3.0 – 3.49 Upper Credit   | B |
| 60 – 69  | B                 | 3                    | 2.5 – 2.99 Lower Credit   | C |
| 50 – 59  | BC                | 2.5                  | 2.0 – 2.49 Pass           | D |
| 40 – 49  | C                 | 2                    | Below 2.0 Fail            | F |
| 30 – 39  | CD                | 1.5                  |                           |   |
| 20 – 29  | D                 | 1                    |                           |   |
| 10 – 19  | E                 | 0.5                  |                           |   |
| 0 – 9    | F                 | 0                    |                           |   |

CGPA Achieved: 3.78/4.00, Distinction

### B.Eng Grading Scale

**Note**

1) **CLASSIFIED DEGREE**

| SCORE (%) | LETTER GRADE | GRADE POINT | CGPA        | CLASS OF DEGREE             |
|-----------|--------------|-------------|-------------|-----------------------------|
| 70 – 100  | A            | 5           | 4.50 - 5.00 | 1 <sup>st</sup> Class       |
| 60 – 69   | B            | 4           | 3.50 - 4.49 | 2 <sup>nd</sup> Class Upper |
| 50 – 49   | C            | 3           | 2.40 - 3.49 | 2 <sup>nd</sup> Class Lower |
| 45 – 49   | D            | 2           | 1.50 - 2.39 | Third Class                 |
| 40 – 44   | E            | 1           | 1.00 - 1.49 | Pass                        |
| 0-39      | F            | 0           | .00 - .99   | Fail                        |

2) **NON-CLASSIFIED DEGREE**  
(To be inserted by the Faculty concerned)

CONSULADO GENERAL DE ESPAÑA EN LAGOS  
Visto Bueno para legalizar la firma y/o el sello de **KINGSLEY ADIABAKPA**  
Ministerio de Asuntos Exteriores de Nigeria por ser, al

CGPA Achieved: 4.77/5.00, First Class

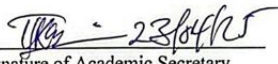
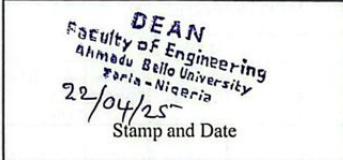
## Signatures and Official Seals

### Head of Department, Prof. Henry Igbadun

|         |                     |  |   |   |   |    |
|---------|---------------------|--|---|---|---|----|
| MMEN523 | Welding And Brazing |  | 2 | A | 5 | 10 |
|---------|---------------------|--|---|---|---|----|

Head of Department, Prof. Henry Igbadun  
Dept. of Agric & Bio-Resource Engineering, ABU, 2025/2026 session

### Dean, Engr. Prof. Jibril Muhammad Kaura & Academic Secretary

|                                                                                     |                                                                                      |                                                                                      |          |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|
| 26/04/2025 25131010005259<br>65994,00 NGN CAYERO                                    | Total Credit Units<br>(Registered)                                                   | ABU Cum Credit Units<br>(Earned)                                                     | ABU CGPA |
|  |  |                                                                                      |          |
| Signature of Dean                                                                   |                                                                                      | Signature of Academic Secretary                                                      |          |
|  |                                                                                      |  |          |
| 22/04/25<br>Stamp and Date                                                          |                                                                                      | Stamp and Date                                                                       |          |

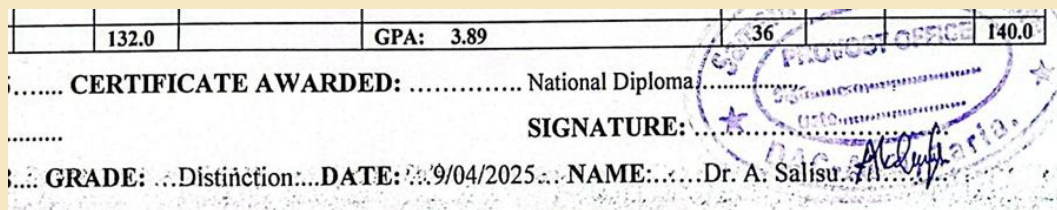
Faculty of Engineering, Ahmadu Bello University, dated 22/04/25

## Federal Ministry of Education & El Cónsul General, José María Ferré Trad



Federal Government of Nigeria & Consulado General de España en Lagos, dated 26 May 2025

### Dr. A. Salisu



Samaru College of Agriculture, ABU, National Diploma certificate, 2016 to 2018

### Course Categories (10 disciplines, full course-by-course marks on the live site)

- Agricultural Machinery, Mechanization & Power (10 courses)
- Soil, Water, Irrigation & Environmental Engineering (18 courses)
- Data Science, Precision Agriculture & Remote Sensing (12 courses)
- Processing, Post-Harvest & Biological Systems (8 courses)
- Mechanical & Materials Engineering (21 courses)
- Electrical, Instrumentation & Structures (10 courses)
- Mathematics, Statistics & Surveying (11 courses)
- Agronomy, Soil Science & Animal Sciences (4 courses)
- Management, Economics & General Studies (28 courses)
- Projects, Practical Exercises & Industrial Training (9 courses)

## PLACES WORKED AND STUDIED

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From Zaria, Nigeria, through Spain (Pamplona, Madrid, Barcelona), Portugal, Greece, Qatar, Morocco, Ethiopia, France, Italy, and Germany, over the course of research travel, conferences, and the Erasmus Mundus programme.



**10 Countries** across fieldwork, study, and the Erasmus Mundus programme.

## RECENT UPDATES

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- Sep 2026: Published GIS monitoring datasets for grape moth and vineyard ripening (Brix), archived on Zenodo with DOIs.
- Aug 2026: Two papers accepted at IJRSI, covering sediment transport modelling and real-time vegetation indexing.
- Aug 2026: Archived STADOS on Zenodo, alongside updated releases of the Zaria irrigation DSS and NaCROP.
- Mar 2025: Paper accepted for presentation at the 6th ARCID, Abuja, on sediment transport rate prediction.

## GALLERY

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Field and campus photographs, organized by theme.

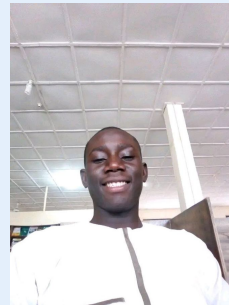
### Graduation Ceremony (3)



**B.Eng graduation ceremony, Class of 2022 cohort (graduated 2024 after COVID-19 and ASUU strike disruptions), Agric & Bio-Resources Engineering, 2019 to 2024**



**Graduation ceremony procession, Bachelor of Engineering, 2019 to 2024**



**National Diploma graduation celebration, Agricultural Engineering Technology, 2016 to 2018**

### EMJMD Programme & Welcome Events (10)



**Large group photo on a stone terrace, EMJMD cohort, 2025**



**Formal welcome event in a university corridor with staff and students, 2025**



**Group photo beside a metal sculpture installation, wind turbines in the distance, 2025**



**Receiving welcome bags on arrival at UTAD, 2026**



**Wearing an EU Global Gateway shirt for an EMJMD event, 2025**



**Outside the Agricultural University of Athens, 2026**



**At the Erasmus+ Pre-Departure Orientation event, 2025**



**In front of the UTAD banner, 2026**



**Beside the official SUSTAGRI  
EMJMD programme banner,  
2025**

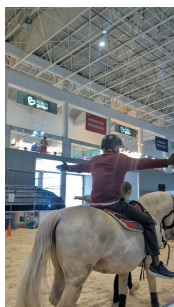


**Group photo with the EMJMD  
SUSTAGRI cohort, 2025**

## **Agricultural Fairs & Machinery Demonstrations (7)**



**Agricultural trade fair, rice and  
grain product stands, 2026**



**Horse-riding demonstration at  
an indoor arena, Forum Braga,  
2026**



**Dairy trade fair with the ALIP  
cow mascot, 2026**



**Farm machinery expo, compact  
tractor on display, 2026**



**Farm machinery expo, standing  
beside tractors, 2026**



**Trying a John Deere  
precision-agriculture GPS  
guidance system, 2026**



**Braga city tourism display at an  
expo, 2026**

## **Douro Valley Vineyard Fieldwork (7)**



**Group visit to Quinta do Seixo vineyard estate, Douro Valley, 2026**



**Visiting the Quinta S. Luiz port wine estate, Douro Valley, 2026**



**Irrigation sensor station on the vineyard terraces, 2026**



**Inside the wine fermentation tank hall, 2026**



**Walking between rows of stainless steel fermentation tanks, 2026**

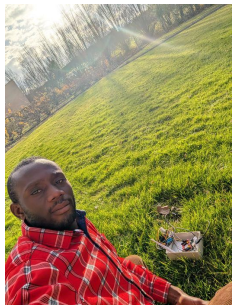


**Inspecting a weather station on the vineyard terraces, 2026**

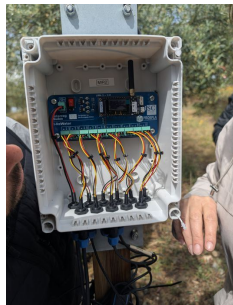


**Fieldwork at the vineyard weather station, 2026**

### **Sensor & Field Research (3)**



**Testing a sensor prototype outdoors on the lawn, 2025**



**Inspecting a LoRa IoT irrigation sensor box in the field, 2025**



**Hyperspectral imaging fieldwork on apples, snow-covered orchard in winter, 2025**

### **Presentations & Conferences (3)**



**Classroom presentation on risks and limitations of spectral sensor deployments, 2025**



**Group selfie with conference lanyards, 2025**



**Presenting at a conference lectern, 2025**

## **Campus Life & Social Gatherings (11)**



**Amigo Secreto (Secret Santa) gift exchange in the classroom, 2025**



**Group celebration after the Secret Santa gift exchange, 2025**



**Evening gathering with classmates at an apartment party, 2025**



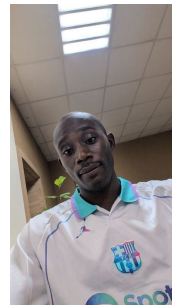
**Bowling and arcade outing with friends, 2025**



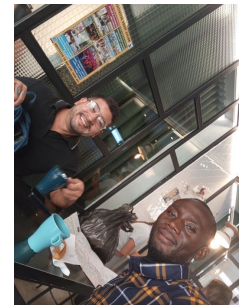
**Group walk near the Public University of Navarre, 2025**



**With friends, football jerseys, outdoor gathering, 2024**



**Casual indoor portrait, 2024**



**Coffee break with a friend on a balcony, 2026**



**Casual indoor portrait, 2025**



**Sharing a meal with a friend in the university cafeteria, 2025**



**Evening gathering with friends in the kitchen, 2025**

## **Portraits & Travel (10)**



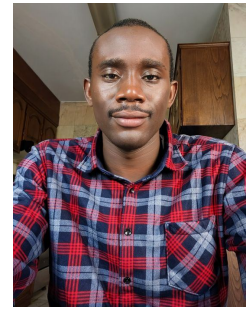
Group hike, resting by a mountain stream, 2025



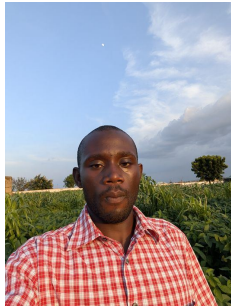
Portrait on an Atlantic coast beach, 2026



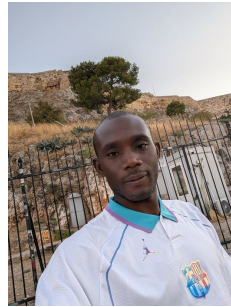
Farm building with a solar panel canopy, 2024



Casual indoor portrait, 2024



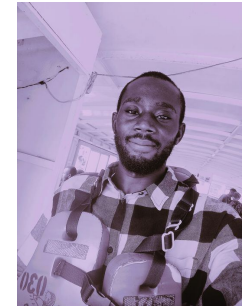
Standing in a maize field at dusk, 2024



Near an ancient fortress wall in Greece, 2026



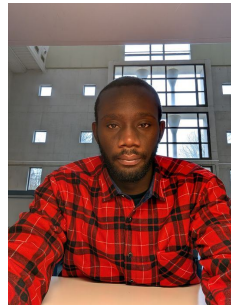
Studying in the university library, 2026



Candid photo with a backpack, purple filter, 2025



Official student portrait for an event, 2025



Studying in a university building, 2026

## Greece — Cape Sounion (4)



Cape Sounion coastline, Greece, 2026



Temple of Poseidon ruins, Cape Sounion, Greece, 2026



Sunset over the Aegean Sea, Cape Sounion, Greece, 2026



Temple of Poseidon columns, Cape Sounion, Greece, 2026

## CONTACT

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Pamplona, Spain | +34 6 5296 7216 | [halilu.175366@e.unavarra.es](mailto:halilu.175366@e.unavarra.es)

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